FACULTY OF ARTIFICIAL INTELLIGENCE

Deep Generative Models

Fourth Year · Course Project

Agentic AI System with Advanced Memory & Multi-Agent Collaboration

COURSE	WEIGHT	TEAM SIZE
Deep Generative Models	**10% of Final Grade**	**7–8 Students**

01
Project Overview

Your team will design and implement a multi-agent AI system grounded in a large language model. The system must orchestrate multiple specialized agents, maintain a persistent memory module, incorporate a retrieval or augmentation mechanism, and expose an interactive user interface. Each team selects their own application domain; the architectural requirements are uniform across all teams.

02
Choosing Your Domain

Select a real-world domain that motivates your system. You are not limited to the examples below — any domain with meaningful data and user queries is valid.

Domain	Example Use Cases
Healthcare	Patient information assistant, symptom checker, medical document Q&A
Education	Course assistant, study planner, exam preparation chatbot
Smart Farming	Crop advice, weather-aware recommendations, agricultural knowledge base
E-Commerce	Product recommendations, order tracking, customer support agent
Legal / Research	Document analysis, case retrieval, summarization assistant

03
Core Requirements

All teams must implement the four components below.

3.1 — Multi-Agent System

The system must be decomposed into multiple specialized agents rather than a single monolithic pipeline. Agents must communicate through defined interfaces, propagate intermediate results across stages, and perform dynamic routing based on the user's input. Inter-agent coordination must be explicit and traceable.

3.2 — Advanced Memory System

The memory module must transcend naive conversation logging. The system must encode, index, and retrieve contextual information such that accumulated knowledge demonstrably enhances response quality over successive interactions.

3.3 — Tool Integration

The system must incorporate a retrieval or augmentation mechanism selected from the approaches below. The choice should be justified by the domain. Teams may compose multiple mechanisms where the application warrants it.

Concept	Full Name	Description
RAG	Retrieval-Augmented Generation	Stores and indexes documents, then retrieves the most relevant pieces at query time to ground the LLM response in real content.
CAG	Cache-Augmented Generation	Pre-computed answers are cached. On repeated or similar queries the system serves from cache without re-running retrieval — improving speed and consistency.
Database Query	Structured Data Retrieval	Converts the meaning of the user's natural language question into a structured query against a relational or other data source, returning formatted results.
Web Search	Live Information Retrieval	Retrieves real-time information from the internet when the query goes beyond what the model or stored documents can answer.

3.4 — User Interface

Each team must deliver a functional interactive interface implemented in Gradio or Streamlit. The interface is a graded component and is not optional.

04
Grading Scheme

The project is worth 10 marks, assessed across five components. An optional bonus is available (see section 5).

#	Component	Criteria	Marks
1	System Architecture	Well-defined agent decomposition, memory design, and module boundaries. Clear interface contracts between components.	2
2	Implementation	All modules operational, integrated, and free of hardcoded logic.	2
3	Agent Collaboration & Intelligence	Coherent inter-agent coordination with dynamic, query-driven decision-making.	2
4	Memory Quality	Accumulated context demonstrably enhances response quality. Advanced memory capabilities implemented.	2
5	User Interface	Functional interactive interface (Gradio or Streamlit).	2
	Total		**10**

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Bonus Features (+2 Marks)

Optional. Awarded only if the base project scores at least 7 out of 10 marks. Teams may attempt one option.

Option	Feature	Requirement
A	Multimodal Input	The system accepts multiple input types (e.g. text and image) and processes them together to generate a unified, coherent response.
B	Multilingual Support	The system handles Arabic and English input and output. Memory and retrieval must function correctly across both languages.
C	Prompt Injection Detection	The routing or task agent validates all user inputs before passing them to the LLM and rejects or flags adversarial inputs.
D	Output Filtering / Guardrails	The response agent inspects and filters generated outputs before they are shown to the user, enforcing safety or relevance constraints.